FUNGI ASSOCIATED WITH TOMATO SPOILAGE FROM RETAIL MARKETS AND MYCOTOXIN PRODUCTION POTENTIAL

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Introduction

Tomato (*Solanum lycopersicum* L.) is one of the most widely consumed vegetables worldwide and represents an important component of the human diet due to its high nutritional value.

Due to their high water activity, soft tissue structure, and nutrient-rich composition, tomatoes represent a favorable substrate for the growth of microorganisms, particularly microscopic filamentous fungi and yeasts.

During cultivation, harvesting, transportation, storage, and retail distribution, tomatoes may become contaminated with fungi originating from soil, air, water, packaging materials, and handling practices. Post-harvest fungal contamination is considered one of the major causes of quality deterioration and economic losses in tomatoes worldwide.

Numerous studies have identified genera such as *Penicillium*, *Alternaria*, *Aspergillus*, *Fusarium*, *Cladosporium*, *Botrytis*, and *Rhizopus* as the most frequent agents of tomato spoilage.

These fungi are capable of colonizing tomatoes through mechanical injuries or natural openings and may proliferate under a wide range of storage conditions, including refrigeration. Beyond visible spoilage, several fungal species associated with tomatoes are of particular concern due to their ability to produce mycotoxins. Mycotoxins are secondary metabolites with well-documented toxic, carcinogenic, mutagenic, and immunosuppressive effects on humans and animals.

The aim of this study was to characterize fungi associated with mould spoilage of retail tomatoes obtained from markets in Slovakia and Hungary, with particular emphasis on the diversity of microscopic filamentous fungi and yeasts and on the ability of selected isolates to produce mycotoxins under *in vitro* conditions.

Material and methods

In the study, 21 tomato samples from Slovak retail chains and 9 tomato samples from Hungarian retail chains were analysed (Tab. 1). Only visibly mouldy tomatoes were included in the study in order to assess the diversity and toxigenic potential of fungi actively involved in spoilage. A piece of mycelium was collected from visibly mouldy parts of each tomato sample and inoculated onto MEA (malt extract agar; HiMedia, India). Subsequently, cultivation was carried out at 25 ± 1 °C for 7 days in a thermostat. Identification at the genus, species, group, or section level was performed based on the study of micro- and macromorphological, physiological, and chemotaxonomic characteristics. The occurence of fungal genera/species/species groups/sections (Frequency of occurence, Fr) was calculated. Selected isolates of potentially toxigenic species (genera *Alternaria, Aspergillus, Fusarium, Penicillium*) were tested by TLC (Thin Layer Chromatography) for their ability to produce selected mycotoxins *in vitro.* The identification of yeasts isolates was carried out using the MALDI-TOF MS system (Bruker Daltonik GmbH, Germany).

Table 1 Analysed samples

No.	Sample	Country of origin	No.	Sample	Country of origin
1S	Cherry tomatoes	Slovakia	16S	Cherry tomatoes	Slovakia
2S	Summer cherry tomatoes	Slovakia	17S	Cherry tomatoes	Slovakia
3S	Cherry tomatoes	Slovakia	18S	Cherry tomatoes	Slovakia
4S	Cherry tomatoes	Slovakia	19S	Cherry tomatoes	Netherlands
5S	Tomatoes	Slovakia	20S	Cherry tomatoes	Tunisia
6S	Tomatoes	Slovakia	21S	Cherry tomatoes	Tunisia
7S	Tomatoes	Slovakia	1H	Tomatoes	Hungary
8S	Tomatoes	Slovakia	2H	Cherry tomatoes	Hungary
9S	Cherry tomatoes	Slovakia	3H	Cherry tomatoes	Hungary
10S	Cherry tomatoes	Netherlands	4H	Tomatoes	Hungary
11S	Cherry tomatoes	Slovakia	5H	Cherry tomatoes	Hungary
12S	Cherry tomatoes	Slovakia	6H	Cocktail tomatoes	Morocco
13S	Tomatoes	Slovakia	7H	Tomatoes	Turkey
14S	Cherry tomatoes	Slovakia	8H	Tomatoes	Turkey
15S	Cherry tomatoes	Tunisia	9H	Tomatoes	Spain

Legend: S – samples from retail chains in Slovakia, H – samples from retail chains in Hungary

Table 4 Toxigenic potential of selected fungal genera

Genus	No. tested	≥1 mycotoxin (%)	Main detected mycotoxins
Penicillium	8	100	PAT, G, RC, CPA, C
Aspergillus	3	33.3	AFB1, AFG1
Alternaria	11	81.8	AOH, AME, ALT
Fusarium	3	33.3	T-2, HT-2, ZON

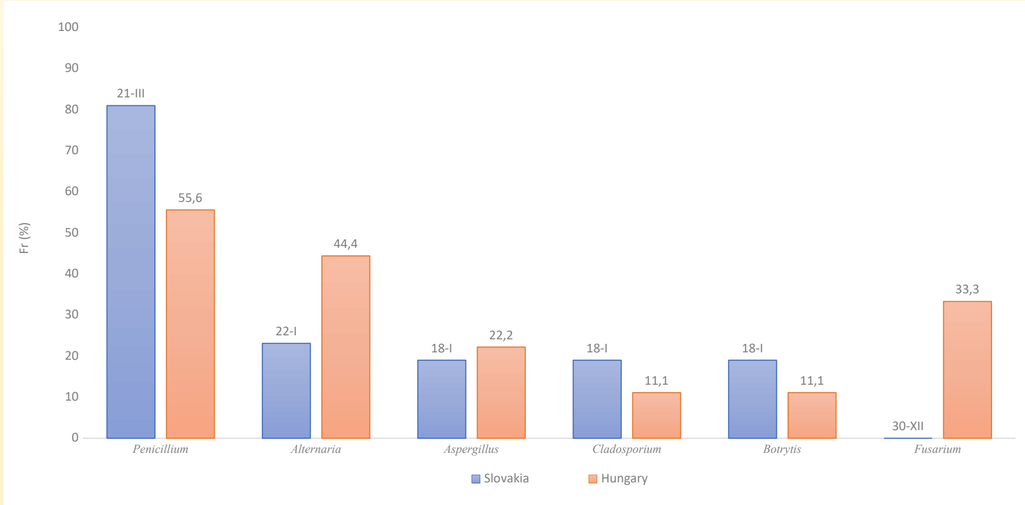
Legend: AFB1 – aflatoxin B_1, AFG1 – aflatoxin G_1, C – citrinin, CPA – cyclopiazonic acid, G – griseofulvin, PAT – patulin, RC – roquefortine C, ALT - altenuene , AME - alternariol monomethylether, AOH - alternariol, ZON – zearalenone

A total of 8 selected isolates of the genus *Penicillium* and 3 isolates of the genus *Aspergillus* were tested for their ability to produce selected mycotoxins by TLC *in vitro* (Tab. 4). Among the five tested isolates of *P. griseofulvum*, all isolates produced griseofulvin, 4 isolates produced patulin, 2 isolates produced roquefortine C, and 2 isolates produced cyclopiazonic acid. The isolate of *P. expansum* produced all tested mycotoxins (roquefortine C, patulin, and citrinin). The isolate of *P. crustosum* produced penitrem A, and roquefortine C, while the isolate of *P. camemberti* produced cyclopiazonic acid. In the case of *Aspergillus,* toxigenic potential was detected only in the isolate belonging to section *Flavi*, which produced aflatoxins B_1 and G_1. *Alternaria* isolates exhibited a high toxigenic potential in the present study, with 81.8% of the tested isolates producing at least one *Alternaria* toxin. Compared to *Penicillium* and *Alternaria, Fusarium* isolates showed a lower toxigenic potential in the present study, with mycotoxin production detected only in one isolate of *F. incarnatum*.

Results

In total, representatives of 9 genera of filamentous fungi as well as yeasts were identified, with their occurence varying among individual samples and between the two countries. The fungal genera detected in the present study, namely *Aspergillus, Alternaria, Botrytis, Cladosporium, Fusarium, Mucor, Penicillium, Rhizopus*, and *Trichoderma*, correspond closely to those most frequently reported in the literature as principal agents of post-harvest tomato spoilage. *Penicillium* was the most frequently detected genus in samples from both countries, confirming its dominant position in the mycocenosis of tomatoes (Fig. 1).

Figure 1 Frequency of occurrence (Fr, %) of dominant genera of microscopic fungi isolated from tomatoes obtained from retail chains in Slovakia (n = 21) and Hungary (n = 9)



Penicillium olsonii was the dominant species, occurring in 17 samples (56.7%). *P. olsonii* was detected in 12 out of 21 samples from Slovak retail chain (57.1%), and in 5 samples from Hungarian markets (55.6%) (Tab. 2). The second most frequently detected species was *P. griseofulvum*, which occurred in 3 samples (10.0%) from Slovak markets. Other species, were detected only sporadically and in Slovak samples only.

Table 2 Occurrence of *Penicillium* species in tomato samples

Species	Slovakia (n = 21)	Hungary (n = 9)	Total isolates	Fr (%)
P. olsonii	12	5	17	56.7
P. griseofulvum	3	0	3	10.0
P. expansum	1	0	1	3.3
P. crustosum	1	0	1	3.3
P. camemberti	1	0	1	3.3
P. sublectaticum	1	0	1	3.3
P. thomii	1	0	1	3.3
P. waksmanii	1	0	1	3.3

The genus *Alternaria* was detected in 9 analysed samples (Fr 30.0%) (Tab. 3). The *Alternaria arborescens* species group was present in 5 samples from Slovak retail chain (Fr 23.8%) and 4 samples from Hungarian markets (Fr 44.4%). The fungal species identified in this study align with findings from authoritative mycological research. Representatives of the genus *Aspergillus* were isolated from only 3 samples (Fr 10.0%) and belonged to three sections: *Circumdati, Flavi*, and *Nigri*. Similarly, species of the genus *Fusarium* were isolated from 3 samples (Fr 10.0%), exclusively from tomatoes obtained from Hungarian retail markets. The identified species included *F. acuminatum*, *F. incarnatum*, and *F. subglutinans*. No *Fusarium* species were isolated from tomatoes originating from Slovak retail chains. The yeast *Geotrichum candidum* was detected in 5 samples (Fr 16.7%).

Table 3 Occurrence of selected fungal species, groups and sections

Taxon / Group	Slovakia	Hungary	Total	Fr (%)
Alternaria arborescens species group	5	4	9	30.0
Aspergillus section *Flavi*	1	0	1	3.3
Aspergillus section *Circumdati*	1	0	1	3.3
Aspergillus section *Nigri*	0	1	1	3.3
Fusarium acuminatum	0	1	1	3.3
Fusarium incarnatum	0	1	1	3.3
Fusarium subglutinans	0	1	1	3.3
Geotrichum candidum	4	1	5	16.7

Acknowledgments

The study was carried out with the financial support of the KEGA 030SPU-4/2024 and VEGA 1/0734/20 project.